

Newman-Penrose Formalism Calculations

Stephani et al., Equation (35.80)

Metric

$$ds^2 = \left(\frac{(3u-8)x^2 + (3u+8)y^2}{4u^3} \right) du \otimes du - du \otimes dv - dv \otimes du + dx \otimes dx + dy \otimes dy$$

Coordinates

$$(x^\mu) = (u, v, x, y)$$

Null Tetrad

Covariant components

$$l = (1) du$$

$$n = \left(-\frac{3x^2}{8u^2} - \frac{3y^2}{8u^2} + \frac{x^2}{u^3} - \frac{y^2}{u^3} \right) du + (1) dv$$

$$m = \left(\frac{1}{2} \sqrt{2} \right) dx + \left(\frac{1}{2} i \sqrt{2} \right) dy$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} \right) dx + \left(-\frac{1}{2} i \sqrt{2} \right) dy$$

Contravariant components

$$l = (-1) dv$$

$$n = (-1) du + \left(-\frac{3x^2}{8u^2} - \frac{3y^2}{8u^2} + \frac{x^2}{u^3} - \frac{y^2}{u^3} \right) dv$$

$$m = \left(\frac{1}{2}\sqrt{2}\right) dx + \left(\frac{1}{2}i\sqrt{2}\right) dy$$

$$\overline{m} = \left(\frac{1}{2}\sqrt{2}\right) dx + \left(-\frac{1}{2}i\sqrt{2}\right) dy$$

Directional Derivatives

$$D = -\partial_v$$

$$\Delta = -\partial_u - \frac{3x^2}{8u^2} - \frac{3y^2}{8u^2} + \frac{x^2}{u^3} - \frac{y^2}{u^3} \partial_v$$

$$\delta = \frac{1}{2}\sqrt{2}\partial_x + \frac{1}{2}i\sqrt{2}\partial_y$$

$$\bar{\delta} = \frac{1}{2}\sqrt{2}\partial_x - \frac{1}{2}i\sqrt{2}\partial_y$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = 0$$

$$\lambda = 0$$

$$\nu = -\frac{3\sqrt{2}x}{8u^2} + \frac{3i\sqrt{2}y}{8u^2} + \frac{\sqrt{2}x}{u^3} + \frac{i\sqrt{2}y}{u^3}$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = 0$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = -\frac{3}{4\,u^2}$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = 0$$

$$\Psi_3 = 0$$

$$\Psi_4 = \frac{2}{u^3}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "N"